

FinSentImpact: A Leakage-Free, Interpretable Re-evaluation of News-Driven Multi-Stock Forecasting

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Abstract—Financial-news sentiment is widely assumed to improve short-horizon stock forecasting, and a recent wave of studies reports near-perfect accuracy (e.g. $R^2 \approx 0.99$) when predicting next-day prices from news and historical data. We show that such results are an artifact of two entangled methodological errors—predicting price levels (which are almost perfectly autocorrelated) and evaluating under a randomly shuffled split—rather than evidence of forecasting skill or of the value of news. Reproducing the standard pipeline on the FNSPID dataset, we obtain $R^2 = 0.999$ for next-day level prediction, but a naive persistence baseline ($\hat{y}_{t+1} = y_t$) attains the same $R^2 = 0.999$, and the identical model applied to next-day returns achieves $R^2 = -0.040$ (worse than predicting zero). We then build a leakage-free, interpretable benchmark over 12 large-cap stocks (2018–2023): stationary features, embargoed chronological walk-forward evaluation, FinBERT and Loughran–McDonald sentiment, strong baselines, and a controlled ablation isolating the marginal value of news. Across five classical models and a GRU, adding news sentiment changes directional accuracy by 0.50 percentage points (95% symbol-clustered CI $[-0.53, 1.60]$)—statistically indistinguishable from zero. We further show that post-hoc attribution can be misleading here: although SHAP and Integrated Gradients assign sentiment a sizable share of in-sample importance (38.3–49.1%), correlation-robust out-of-sample tests—grouped block-permutation (0.48 pp accuracy change), the sentiment-only ablation, and Diebold–Mariano—show this corresponds to no predictive value. We treat the data’s severe multicollinearity (VIF up to 29,282), temporal autocorrelation, and cross-sectional correlation as first-class threats and address each. Our results temper optimistic claims about news-driven forecasting and provide a fully reproducible protocol and artifact for honest evaluation.

Index Terms—financial forecasting, news sentiment, data leakage, reproducibility, FinBERT, explainable AI, FNSPID

I. Introduction

Predicting stock movements from text is a long-standing goal of financial machine learning [1]–[3]. The release of large, time-aligned news corpora such as FNSPID [4] has renewed interest in multimodal pipelines that fuse news sentiment with historical prices, and a number of recent works report strikingly high accuracy on next-day prediction. Headline figures of $R^2 \approx 0.99$ are common when the target is the next day’s price level.

We argue that these numbers are, with high probability, illusory. Two errors recur. First, predicting price levels

exploits the near-unit-root autocorrelation of prices: because $\text{corr}(P_t, P_{t+1}) \approx 0.999$, a model that merely echoes today’s price scores $R^2 \approx 0.99$ without any predictive content. Second, random train/test splitting of a time series scatters each day’s temporal neighbours across the train and test sets, letting the model interpolate rather than forecast and leaking future information. Neither error is exotic; both are easy to commit and hard to notice, because the resulting metrics look excellent.

This paper makes the following contributions:

- A concrete reproduction and refutation. On FNSPID we reproduce $R^2 = 0.999$ for next-day levels, then show a persistence baseline matches it ($R^2 = 0.999$) and that the same model on returns yields $R^2 = -0.040$. The apparent skill is autocorrelation (Section V).
- A leakage-free, interpretable benchmark over 12 large-cap stocks with stationary features, embargoed chronological walk-forward evaluation, strong baselines (persistence, historical mean, majority class), and modern sentiment (FinBERT, Loughran–McDonald).
- A controlled sentiment ablation with dependence-aware inference (symbol-clustered bootstrap, HAC Diebold–Mariano) showing that engineered news sentiment adds no statistically reliable value for next-day return or direction prediction.
- A four-method explainability analysis (SHAP, permutation, ALE, Integrated Gradients) that quantifies—rather than merely displays—how little sentiment contributes, and an explicit treatment of the severe multicollinearity that undermines naive attribution.
- A reproducible artifact: a Spark-based ingestion pipeline, fixed splits, and one-command reproduction on the public FNSPID data.

Our aim is constructive. Text can matter for markets; but establishing that requires evaluation protocols that cannot be satisfied by autocorrelation alone. We provide such a protocol and a baseline against which genuinely news-driven gains can be measured.

II. Related Work

Text-driven forecasting. Event- and sentiment-based models have a long history [1]–[3], more recently using domain-specific encoders such as FinBERT [5] and cross-modal fusion architectures [6]. FNSPID [4] provides large-scale, time-aligned news and finds that sentiment gains are modest and model-dependent—consistent with our findings.

Leakage and evaluation. The dangers of look-ahead bias and improper cross-validation in finance are well documented [7]; purged, embargoed, walk-forward protocols are recommended precisely because adjacent observations are serially correlated. The efficient-market view [8] further predicts that next-day returns are close to unforecastable from public information, making strong naive baselines essential. Robustness and consistency testing of financial NLP models is likewise emphasized by FinTrust [9].

Explainability. SHAP [10] and LIME [11] are standard but can misattribute importance under correlated features; Accumulated Local Effects (ALE) [12] were designed to remain valid in that regime, and Integrated Gradients [13] provide axiomatic attribution for neural models. We use all four, together with permutation importance, to triangulate.

III. Data and Pipeline

Problem setup (what is X , what is Y). This is a supervised time-series problem, and we state it precisely because the entangling of X and Y across time is the source of the errors we study. For each stock s and trading day t we build a feature vector $\mathbf{x}_{s,t}$ using only information available at the close of day t —lagged returns, rolling volatility, RSI, momentum, moving-average gaps, range/volume statistics, and news-sentiment features aggregated from articles known by that close. The target is realized on the next trading day:

$$y_{s,t}^{\text{ret}} = \log \frac{P_{s,t+1}}{P_{s,t}}, \quad y_{s,t}^{\text{dir}} = \mathbb{I}[y_{s,t}^{\text{ret}} > 0],$$

with the price level $P_{s,t+1}$ used only as a diagnostic target in the leakage study. Every feature is strictly backward-looking (verified by an empirical leakage screen, Sec. IV), targets are produced by a per-ticker forward shift, and all splitting preserves temporal order. We never treat the panel as i.i.d. tabular data.

Sources. We use FNSPID [4]: per-ticker daily OHLCV and an external financial-news corpus. Because the news file is multi-gigabyte, ingestion and filtering use Apache Spark [14] (streaming, bounded-memory), after which the compact per-ticker panel is processed in pandas. We study 12 large-cap stocks with adequate news coverage over 2018–2023, yielding 17,790 stock-day observations and 18,600 aligned articles (news present on 56.7% of stock-days).

Time-aware alignment (no leakage). Each article is timestamped in US/Eastern and mapped to a trading

session so that only information available by the close of day t can inform the prediction of day $t+1$: intraday news maps to the same session, after-close and non-trading-day news to the next session. Sentiment is aggregated per (stock, session).

Sentiment. We replace opaque or proprietary sentiment scores with two reproducible signals: (i) FinBERT [5] probabilities (positive–negative), and (ii) the Loughran–McDonald finance lexicon [15] polarity. Both are aggregated to daily features with short rolling histories and a news-count/volume signal.

Features and targets. All predictive features use only information available at the close of day t : lagged returns, rolling volatility, RSI, momentum, moving-average gaps, range/volume statistics, and the sentiment features. Targets are next-day log-return r_{t+1} , direction $\mathbb{I}[r_{t+1} > 0]$, and (for the leakage study only) the next-day price level.

IV. Exploratory Data Analysis and Data Integrity

Before modelling we run a time-series EDA that both validates the setup and surfaces a data-integrity problem that would otherwise corrupt every return-based result.

Non-stationary levels, stationary returns. Augmented Dickey–Fuller tests find 100% of the price-level series non-stationary but 100% of the return series stationary; lag-1 autocorrelation is 0.987 for levels versus -0.108 for returns (Fig. 1). This is the formal justification for predicting returns rather than levels—the apparent “predictability” of levels is a unit-root artifact.

Data integrity (a senior-analyst catch). A first EDA pass showed an impossible excess kurtosis (> 4000) and no volatility clustering—red flags for real equity returns. The cause was a handful of impossible single-day returns (e.g. AMZN -294% on 2020-07-06) where FNSPID splices unadjusted and split-adjusted prices. We compute returns from adjusted closes and mask $|r| > 0.5$ (a data error for these large caps, not a market move). After cleaning, excess kurtosis is a realistic 12.7 with pronounced volatility clustering (Ljung–Box on squared returns, $p < 10^{-3}$), matching known stylized facts (Fig. 4).

Cross-sectional dependence. The twelve stocks co-move strongly (mean pairwise return correlation 0.41; Fig. 3), so stock-days are far from independent and our inference clusters by stock.

Why the task is hard, and a leakage screen. The largest absolute correlation between any backward-looking feature and the next-day return is only 0.060 (sentiment peaks at 0.016; Fig. 2). This near-orthogonality is the honest reason next-day prediction is near a coin flip, and it doubles as a leakage screen: no feature is suspiciously predictive, as look-ahead would produce.

V. The Illusion of Skill: A Leakage Study

We first reproduce the standard “high-accuracy” setup and dissect it. Table I and Fig. 5 report four experiments with the same Random Forest.

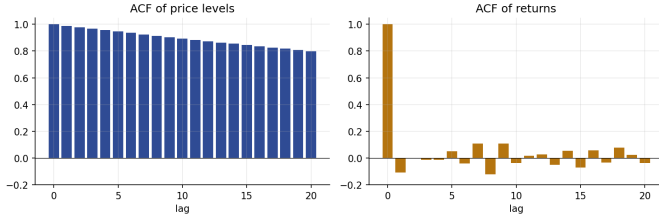


Fig. 1. Autocorrelation: price levels (left) are near-unit-root; returns (right) are effectively white. Predicting levels exploits the left panel.

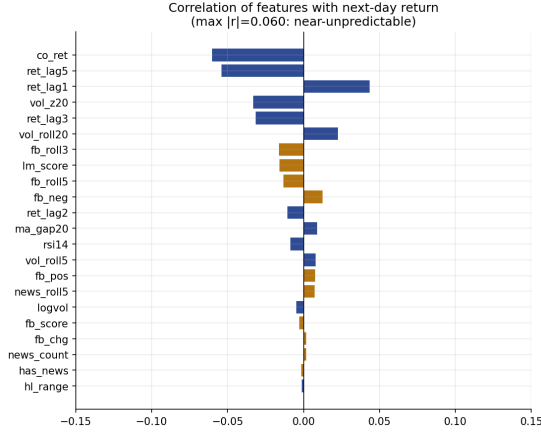


Fig. 2. Correlation of each backward-looking feature with the next-day return. The maximum is 0.060; sentiment features (orange) are near zero.

The random-split level model attains $R^2 = 0.999$; keeping the same features and target but using a chronological split barely changes it ($R^2 = 0.998$), because the “skill” is autocorrelation, not the split. The decisive comparison is the persistence baseline $\hat{P}_{t+1} = P_t$, which attains $R^2 = 0.999$ —indistinguishable from the model. When the same model instead predicts returns, performance falls to $R^2 = -0.040$, i.e. worse than predicting a constant zero. Level-scale R^2 is therefore not a meaningful measure of forecasting skill here.

VI. Correlation Structure of the Data

A second hazard—and the one most likely to threaten conclusions in this literature—is the strong correlation among features, across stocks, and across time. We treat it as a first-class threat to validity.

Three kinds of correlation. (i) Feature multicollinearity: Fig. 6 reports variance inflation factors (VIF); the raw OHLCV levels used by prior pipelines are near-perfectly collinear (VIF up to 29,282, versus the usual threshold of 10), which destabilizes linear coefficients and invalidates PDP- and coefficient-based explanations. (ii) Temporal autocorrelation: price levels are near-unit-root and overlapping features are serially correlated, so IID cross-validation and IID significance tests are invalid. (iii) Cross-sectional correlation: the stocks co-move with the

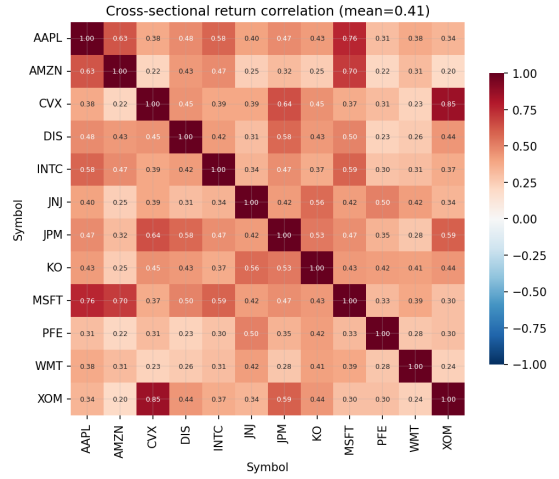


Fig. 3. Cross-sectional return correlation across the twelve stocks (mean 0.41): observations are not independent, motivating symbol-clustered inference.

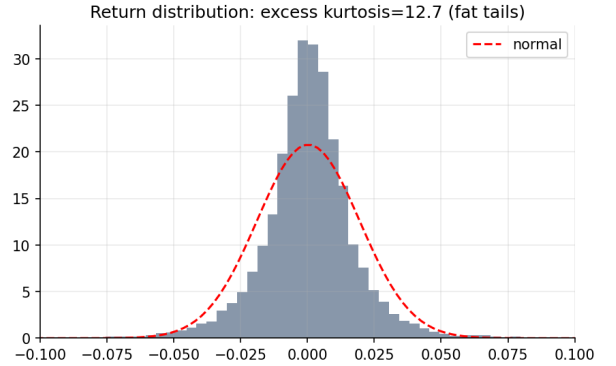


Fig. 4. Return distribution after data cleaning: fat-tailed (excess kurtosis 12.7) but free of the splice artifacts that originally produced kurtosis > 4000 .

market, so pooled stock-day observations are far from independent and naive tests overstate significance.

Why our conclusion is robust to all three. Crucially, our central claim is a null (news sentiment adds no measurable value). Correlation threatens a null in only one way—by masking a real effect behind correlated price features—and we rule this out directly: the sentiment-only models (Sec. VII-A) test sentiment in complete isolation and still achieve no skill ($R^2 \leq 0$, $\text{MCC} \approx 0$), so no price feature can be hiding it. The other failure mode of correlation—inflating apparent significance—can only make a positive claim look stronger, and we make no positive claim; we nonetheless guard against it so our “no-effect” confidence intervals are honest. Concretely we use: stationary transforms (returns, ratios, z -scores; max VIF 5, a $\sim 17\times$ reduction); ALE [12] and grouped attribution rather than PDP or per-feature bars; grouped block-permutation importance that permutes the entire sentiment block jointly (preserving within-block correla-

TABLE I

Next-day prediction under different protocols (Random Forest).

Random-split level prediction reproduces $R^2 \approx 0.99$, but a persistence baseline matches it and the honest return task collapses.

Protocol	Target	R^2	RMSE
RF, random split	level	0.999	7.223
RF, chronological	level	0.998	2.819
Persistence $\hat{P}_{t+1}=P_t$	level	0.999	2.241
RF, chronological	return	-0.040	0.017

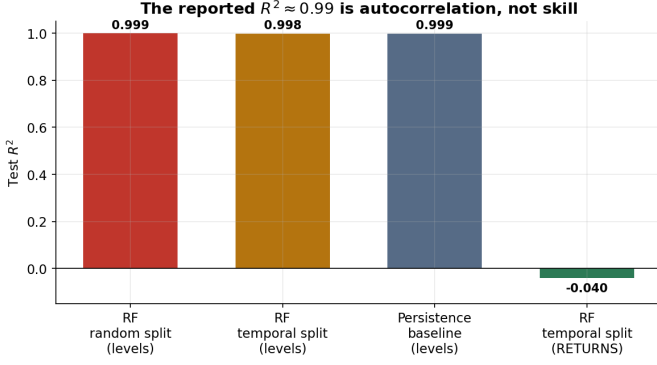


Fig. 5. The reported $R^2 \approx 0.99$ is autocorrelation, not skill. A naive persistence baseline equals the model on levels; predicting returns yields $R^2 < 0$.

tion, so redundancy cannot hide the block’s contribution); embargoed walk-forward splits; a symbol-clustered bootstrap that resamples whole stocks (respecting cross-sectional dependence); and Newey–West HAC Diebold–Mariano tests (respecting temporal dependence). Every significance statement below uses stocks, not stock-days, as the effective unit.

VII. Leakage-Free Evaluation

Protocol. We use an expanding-window walk-forward: for each test year $Y \in \{2021, 2022, 2023\}$ we train on all data before Y (with a 10-day embargo) and test on Y . Features are standardized on the training fold only. We report per-fold metrics and their mean.

Models and baselines. Ridge/Logistic, Random Forest, XGBoost, LightGBM, SVM, and a two-layer GRU. Baselines: persistence and historical-mean returns (regression) and majority-class/always-up (direction).

Ablation. Each model is trained on price-only, sentiment-only, and price+sentiment feature sets, isolating the marginal contribution of news.

A. Results

Table II summarizes the honest tasks. On next-day returns, every model’s R^2 is near zero or negative—none beats the persistence or historical-mean baseline (Fig. 7). On direction, the best model reaches 50.8% accuracy versus a 51.3% majority baseline, with Matthews correlation near zero (Fig. 8). Performance is weak but stable across regimes, including the 2022 drawdown (Fig. 9).

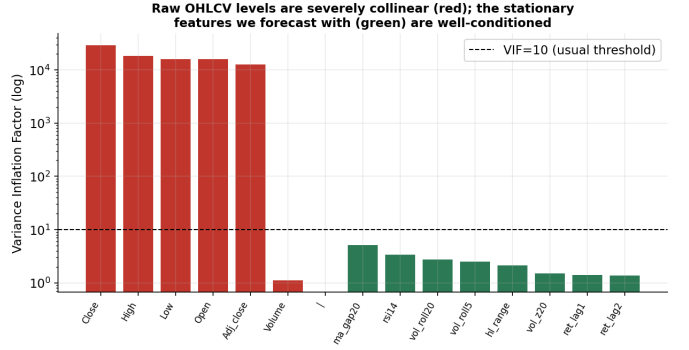


Fig. 6. Variance inflation factors. Raw OHLCV levels (red) are severely collinear; the stationary features we use (green) are well-conditioned.

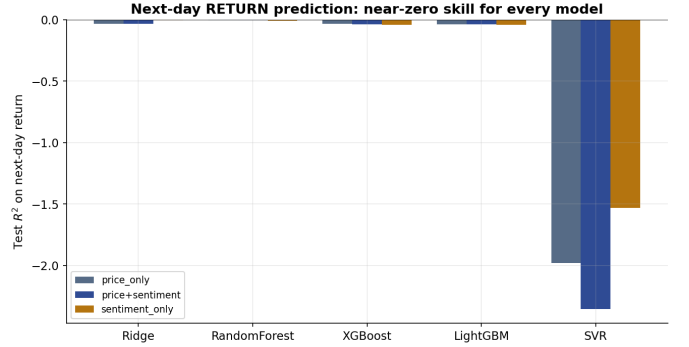


Fig. 7. Next-day return R^2 by model and feature set. All near zero; sentiment does not shift the picture.

Importantly, the sentiment-only models—which isolate news from all price information—are no better than chance, establishing that sentiment carries no standalone next-day signal here and that its non-contribution in the combined model is not an artifact of price features masking it.

B. Does news sentiment help? A dependence-aware test

Table III reports the marginal value of sentiment. Adding news features changes directional accuracy by 0.50 percentage points, with a 95% symbol-clustered bootstrap CI of $[-0.53, 1.60]$ that straddles zero; Wilcoxon tests across symbols are non-significant. For returns, HAC Diebold–Mariano tests find no consistent reduction in forecast error (Fig. 10).

VIII. Explainability: Attribution Is Not Predictive Value

Applying explainability here yields a cautionary lesson rather than a simple verdict, and interpreting it carefully matters. In-sample attribution assigns news sentiment a sizable role: SHAP gives the ten sentiment features 38.3% of total importance (Fig. 11) and the GRU’s Integrated Gradients give sentiment channels 49.1% (Fig. 13). Read naively, this looks like evidence that sentiment matters.

TABLE II

Leakage-free walk-forward results (mean over test years 2021–2023). No model beats the naive baselines on returns; directional accuracy is near chance; adding sentiment does not help.

Model	Return (R^2 / RMSE / DirAcc)			Direction (Acc / F1 / MCC)		
Baselines						
persistence/zero	-0.002	0.017	—	—	—	—
hist. mean / majority	-0.002	0.017	—	0.513	0.678	0.000
Ridge (price)	-0.034	0.017	0.491	0.499	0.574	-0.009
Ridge (+sent)	-0.035	0.017	0.495	0.497	0.540	-0.009
RandomForest (price)	-0.005	0.017	0.513	0.496	0.485	-0.005
RandomForest (+sent)	-0.005	0.017	0.508	0.502	0.474	0.008
XGBoost (price)	-0.033	0.017	0.497	0.499	0.554	-0.008
XGBoost (+sent)	-0.035	0.017	0.498	0.504	0.550	0.004
LightGBM (price)	-0.037	0.017	0.500	0.500	0.553	-0.005
LightGBM (+sent)	-0.037	0.017	0.491	0.503	0.551	0.003
SVR (price)	-1.982	0.029	0.489	0.508	0.591	0.005
SVR (+sent)	-2.354	0.030	0.489	0.495	0.562	-0.016

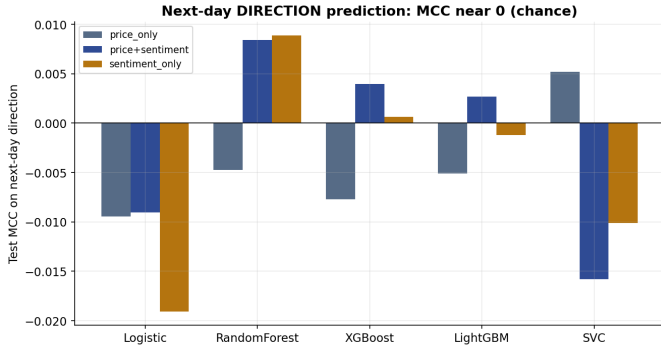


Fig. 8. Directional MCC by model and feature set. Near-zero (chance) for all.

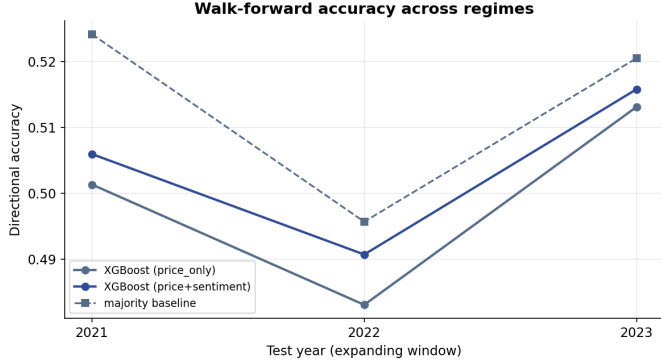


Fig. 9. Walk-forward directional accuracy across regimes (2021–2023).

It is not. Attribution measures how a flexible model uses features to fit the training data, not their out-of-sample predictive value—and the two diverge sharply here. The correlation-robust grouped block-permutation test, which permutes the entire sentiment block jointly so

TABLE III

Marginal value of news sentiment (price+sentiment vs. price-only), with dependence-aware inference.

Model	ΔAcc (pp)	95% CI	Wilcoxon p	DM t (HAC)
Logistic	-0.26	$[-0.86, 0.44]$	0.38	—
XGBoost	0.50	$[-0.53, 1.60]$	0.62	-0.30
LightGBM	0.34	$[-0.65, 1.26]$	0.47	0.03

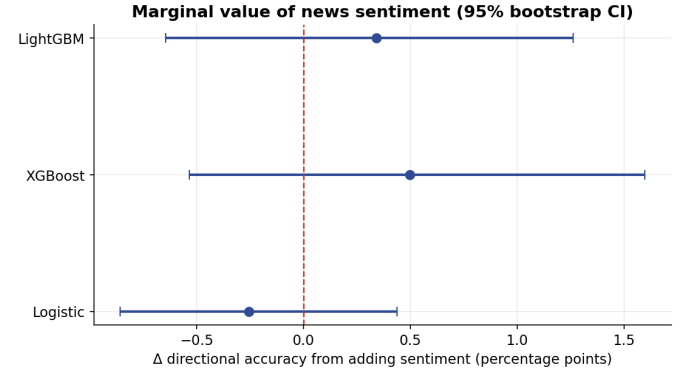


Fig. 10. Change in directional accuracy from adding sentiment, with 95% symbol-clustered bootstrap CIs. All intervals include zero.

within-group redundancy cannot hide a real effect, changes test accuracy by only 0.48 points—within noise, and no larger than for the price block (0.45 points) on this near-unpredictable task (Fig. 12). ALE curves for sentiment features are nearly flat, and the sentiment-only models (Sec. VII-A) have no skill. Thus the large SHAP/IG share reflects in-sample fitting of noise, not signal. The methodological takeaway, relevant to the wider use of XAI in finance, is that a large attribution share is not evidence of predictive value; the ablation and block-permutation tests, not the attribution bars, settle the question.

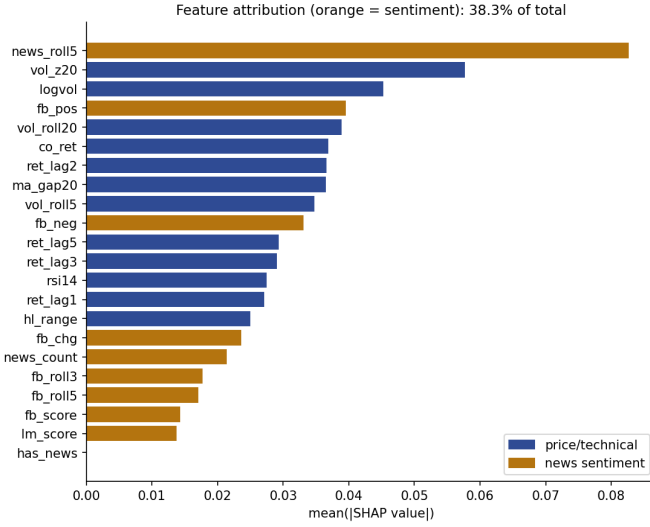


Fig. 11. Grouped SHAP attribution (orange: news sentiment). Sentiment receives a sizable in-sample share (38.3%)—which, as Fig. 12 shows, does not translate to out-of-sample value.

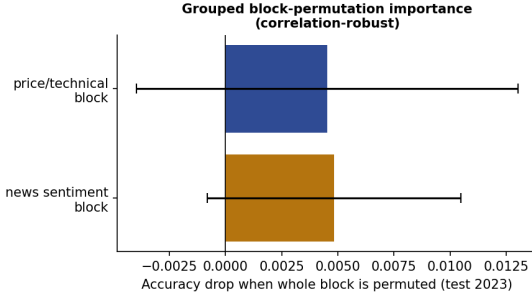


Fig. 12. Correlation-robust grouped block-permutation. Jointly permuting the news sentiment block changes accuracy by only 0.48 pp (within noise), despite sentiment’s large SHAP share; next-day direction is near-unpredictable from either feature group.

IX. Discussion and Limitations

Careful interpretation. We are deliberately precise about what our null means. We do not claim that news sentiment has exactly zero effect; we claim that under leakage-free, dependence-aware evaluation there is no statistically detectable marginal value of engineered daily sentiment for next-day prediction of these large, liquid stocks. The confidence intervals include zero and are consistent with, at most, a tiny effect. Equally, the sizable SHAP/IG attribution to sentiment (Sec. VIII) should not be read as evidence of value: it is in-sample fitting, contradicted by every out-of-sample test.

Threats to validity. The chief threat in this domain is correlation, and we address its three forms directly (Sec. VI): feature multicollinearity (mitigated by stationary features and ALE/grouped attribution), temporal autocorrelation (embargoed walk-forward; HAC Diebold–Mariano), and cross-sectional co-movement (symbol-clustered inference). Because our central claim is a null,

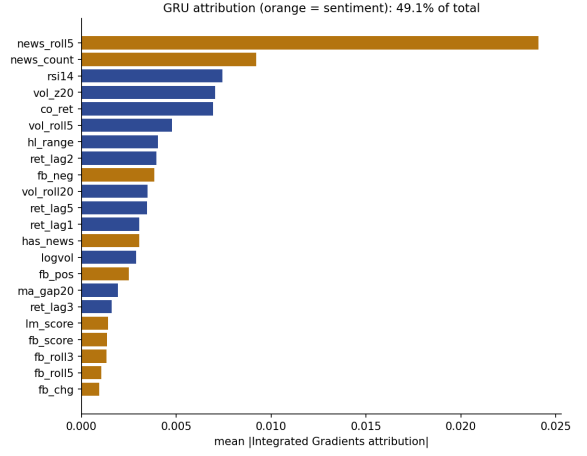


Fig. 13. GRU Integrated-Gradients attribution. Sentiment channels (orange) get a large in-sample share, again not reflected in out-of-sample performance.

the remaining risk is that correlated price features mask a real sentiment effect—which the sentiment-only models and grouped block-permutation rule out. A second threat, data integrity, was caught by EDA and fixed (Sec. IV).

Scope. The study covers 12 large caps with adequate FNSPID coverage (56.7% of days), daily granularity, and headline-level sentiment. Text may still help at intraday or multi-day horizons, for less-liquid names, around scheduled events, or via richer representations than aggregate sentiment; we make no claim beyond our regime. The lasting contribution is the protocol: any claim of news-driven skill should first clear persistence and historical-mean baselines on returns, under embargoed walk-forward evaluation, with dependence-aware inference.

X. Conclusion

Reported near-perfect accuracy in news-driven stock forecasting can arise entirely from price autocorrelation and improper splitting. Replacing level prediction with returns, random splits with embargoed walk-forward, and naive metrics with strong baselines and dependence-aware tests, we find that engineered news sentiment does not measurably improve next-day forecasting for large-cap stocks, and four explainability methods confirm its minor role. We release a reproducible, interpretable benchmark so that future claims of news-driven skill can be evaluated honestly.

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